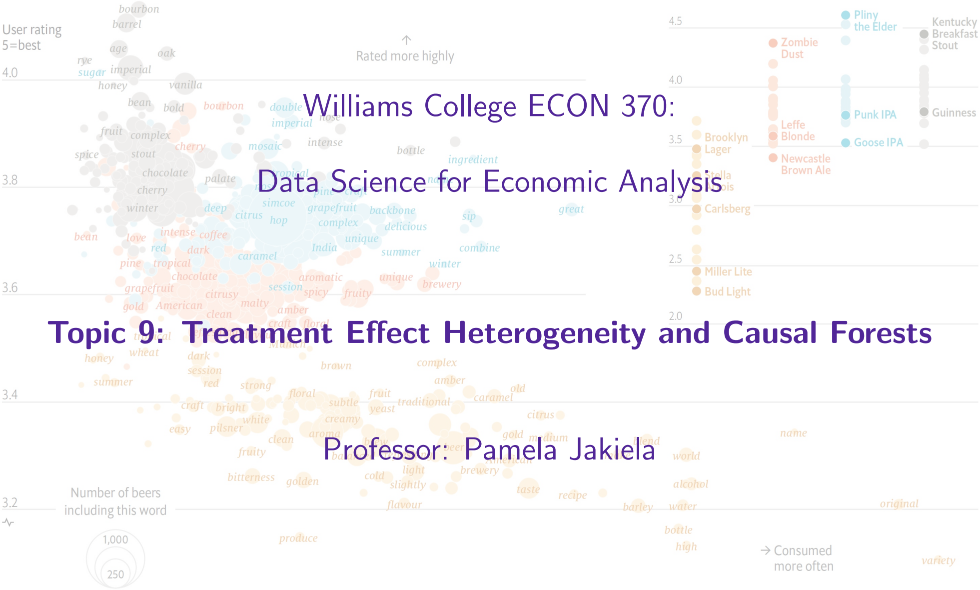

Williams College ECON 370:

Data Science for Economic Analysis

Topic 9: Treatment Effect Heterogeneity and Causal Forests

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Potential Outcomes and Treatment Effects

We are interested in the impact of treatment, and each individual is either treated or not:

- $D_i =$ is a **treatment dummy** equal to 1 if i is treated and 0 otherwise

For each individual i, there are two **potential outcomes**:

- $Y_{0,i} = i$'s outcome if she **doesn't** receive treatment
- $Y_{1,i} = i$'s outcome if she **does** receive treatment

The **causal impact** of treatment on individual i is $\delta_i := Y_{1,i} - Y_{0,i}$

- We are interested in estimating $E[\delta|\text{some group of eligible units}]$
  - The eligible population, those who take up treatment, some subgroup of eligible units

Random Assignment

The fundamental problem of causal inference: for each individual, we observe

$$\begin{aligned} Y_i &= \underbrace{(Y_{1,i} - Y_{0,i})}_{=\delta_i} D_i + Y_{0,i} \\ &= \delta_i D_i + Y_{0,i} \end{aligned}$$

When treatment is randomly assigned, the treatment and control groups are random samples of the underlying population of eligible units (people, classrooms, firms, communities, etc.)

$\Rightarrow \; E[Y_0|D_i = 1] = E[Y_0|D_i = 0] = E[Y_0]$

$\Rightarrow \; E[Y_1|D_i = 1] = E[Y_1|D_i = 0] = E[Y_1]$

Regression Estimates of the Average Treatment Effect (ATE)

Estimate the OLS regression: $Y = \alpha + \beta D$

- $\hat{\alpha} = \bar{Y}_C$, the average outcome in the control group
- $\hat{\beta} = \bar{Y}_T - \bar{Y}_C$, the difference in means between the treatment and control groups

$\hat{\beta}$ provides an unbiased estimator of the average treatment effect

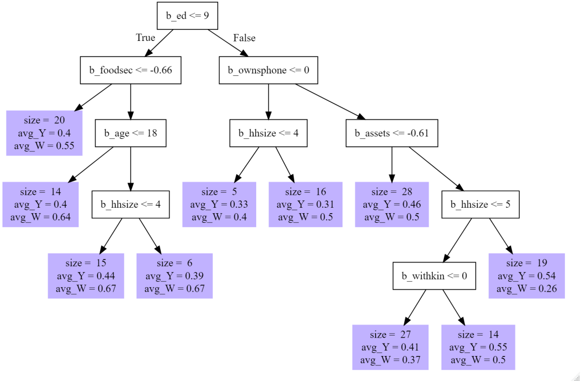
$$\begin{aligned} E[\hat{\beta}] &= E[\bar{Y}_T - \bar{Y}_C] \\ &= E[\bar{Y}_T] - E[\bar{Y}_C] \\ &\dots \\ &= E[\delta_i] \end{aligned}$$

because $E[Y_0|D_i = 1] = E[Y_0|D_i = 0] = E[Y_0]$ when treatment is randomly assigned

3 Observations About ATEs in RCTs

1. Same holds (in expectation) in any subsample defined in terms of baseline covariates
   - $E[\bar{Y}_T|X = x] - E[\bar{Y}_C|X = x]$ provides an estimate of ATE in group with $X = x$
2. When $Y_{0,i} = 0$ for all i, we can use $Y_{1,i}$ to estimate the treatment effect
   - Again, $E[\bar{Y}_C|X = x]$ provides an estimate of ATE in group with $X = x$
3. If we can estimate $Y_{1,i}$ and $Y_{0,i}$ (well), we can estimate δ_i

Causal Forests



A **causal forest** is a machine learning algorithm for identifying treatment effect heterogeneity (Athey and Imbens 2016, Davis and Heller 2017, Wager and Athey 2018)

Causal Forests Are Built from Regression Trees

A **causal forest** is a machine learning algorithm for identifying treatment effect heterogeneity

- A **causal tree** is a regression tree where successive partitions are chosen to minimize expected mean squared error of estimated treatment effects (instead of MSE of Y)
  - Identifying sub-samples with less variation in treatment effect
  - In practice, this involves maximizing the sum of the squared leaf-specific estimates of the treatment effect (which are the differences between the treatment and control means)
- A **causal forest** is a random forest of causal trees
- **Honest trees** use split sample methods for partitioning, estimation, inference
  - Standard test vs. training data splits use the same data to build tree and make predictions
  - Allows for the construction of standard errors

Causal Forests in R and Python

- Causal forests are implemented as double ML (relevant when treatment is not random)
  - Use X variables to predict outcome Y and treatment W
  - Train the causal forest on residualized $\tilde{Y} = Y - \hat{Y}$ and $\tilde{W} = W - \hat{W}$
    - In R (`grf`), we can set the predicted values of to 0 or the mean (in an RCT)
    - In Python (`econml`), we have to choose an ML model to predict Y, W
  - Tuning parameters are similar to regression trees and random forests
    - Number of trees, minimum leaf size, depth (Python)
    - Proportion treated within a leaf is also relevant with trees

Interpreting Causal Forests

Causal forests (can) tell us two things:

- Is there (meaningful) treatment effect heterogeneity that is explained by observables?
  - Machine learning is noisy in small samples!
  - Algorithm looks for heterogeneity that is predicted by covariates
- Which baseline covariates predict the observed treatment effect heterogeneity?

Relevant outputs from causal forests:

- What are the observation-level out-of-bag (OOB) predicted conditional ATEs (CATEs)?
- Does the variation in OOB CATES explain variation in Y (test calibration in R)?
- Are estimated treatment effects higher for observations w/ high OOB CATEs?
- How often was a covariate was used to split tree?

Example: EMERGE Data

Storybook Comprehension Questions Capture Familiarity with Story

What 3 things did Caro forget at home today?

What do you think Meg and Ben are looking at?

EMERGE Impacts on Children's Storybook Comprehension

Survey:	**Midline**	**Endline**
Sample:	**All Children**	**All Children**
Luo books treatment	4.939	5.969
	(0.390)	(0.138)
	[$p < 0.001$]	[$p < 0.001$]
English books treatment	4.339	5.468
	(0.257)	(0.104)
	[$p < 0.001$]	[$p < 0.001$]
Control group mean	0.420	0.772
H_0: Luo books = English books	0.090	$p < 0.001$
Observations	357	4464
R^2	0.655	0.678

Treatment Effects Vary by Age

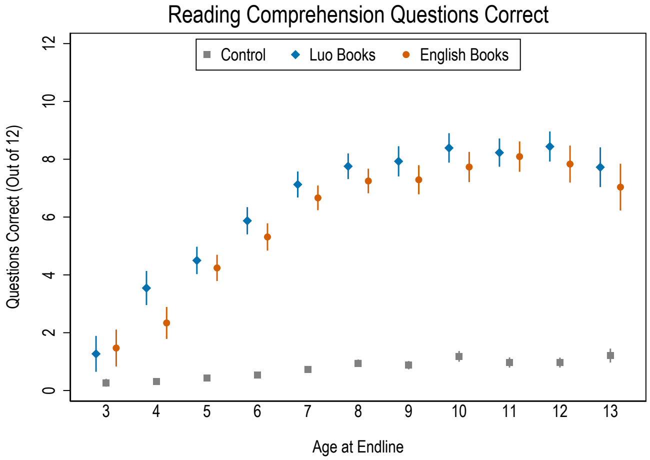
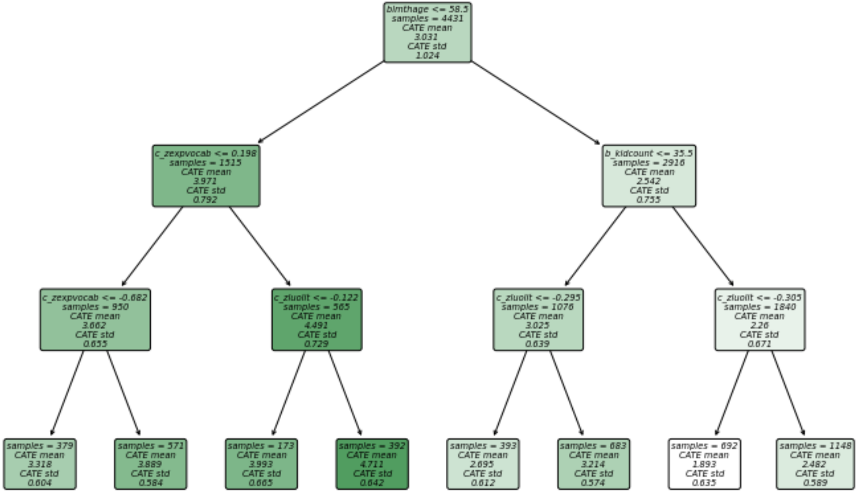
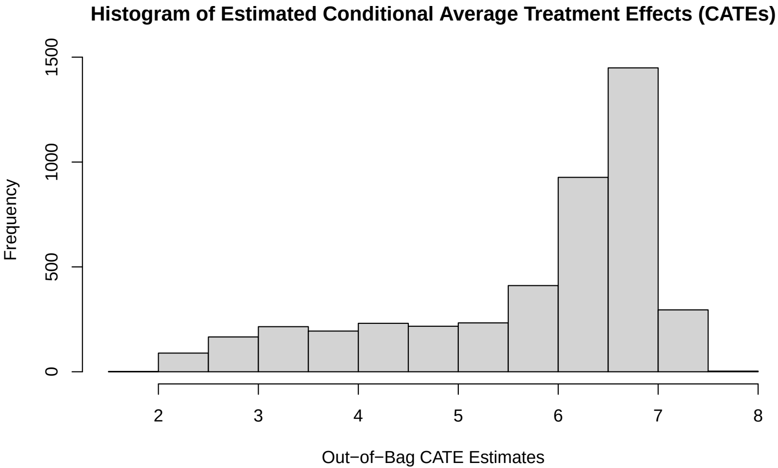


Illustration of a Single Causal Tree



Causal Forest Example



Is There (Statistically Significant) Treatment Effect Heterogeneity?

Partition sample into quantiles based on OOB CATE estimates

- Honest estimation approach allows for estimate of ATE, SE in each CATE quantile
- Is ATE in highest quantile (significantly) different from ATE in lowest quantile?

Alternative approach: do CATE estimates explain variation in Y?

- Generate two variables: $\bar{\tau}$ (ATE) and τ_i (individual CATE estimate)
- Regress Y on $\bar{\tau} D_i$ and $(\tau_i - \bar{\tau})\, D_i$
- Coefficient on $(\tau_i - \bar{\tau})\, D_i$ tests the hypothesis that variation in individual-level predicted CATEs explains variation in the outcome in the treatment group (relative to control)

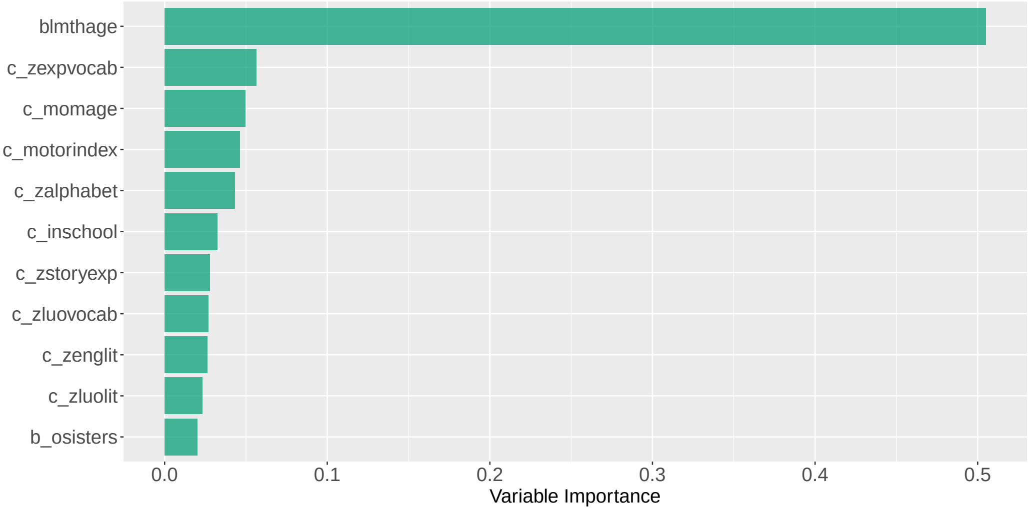
Testing for Treatment Effect Heterogeneity: Example

```
cf_test <- test_calibration(cf)
cf_test
```

```
Best linear fit using forest predictions (on held-out data)
as well as the mean forest prediction as regressors, along
with one-sided heteroskedasticity-robust (HC3) SEs:

                               Estimate Std. Error t value Pr(>t)
mean.forest.prediction         0.99552 0.16848 5.9089 1.851e-09 ***
differential.forest.prediction 1.17797 0.22644 5.2021 1.030e-07 ***
---
Signif.  codes:  0 *** 0.001 ** 0.01 * 0.05 .  1
```

Variable Importance in Causal Forests: Example



Generic Machine Learning Approach

Test for heterogeneous treatment effects following Chernozhukov et al. (2024)

- Propose a **generic machine learning** approach allows for inference about **key features** of treatment effect heterogeneity, can be used with any (or many) machine learner(s)
- Use machine learning to predict potential outcomes with and without treatment, and hence implicitly predict the treatment effect as a function of baseline characteristics
- Use predictions to:
  1. Test for (predictable) treatment effect heterogeneity
  2. Test for differences in impact between most-impacted and least-impacted groups
  3. Identify characteristics of most-impacted and least-impacted groups (not today)
  4. Generate individual-level proxies for estimated **conditional average treatment effects**

Generic Machine Learning: Setup

Approach requires the following:

- Outcome: Y
- Baseline characteristics: Z
- Randomly-assigned treatment: D

Split the sample, use any ML approach to estimate (auxiliary data) and predict (main data):

- $b(Z)$: proxy for the **baseline conditional average**: $b_0(Z) = E[Y_0|Z]$
- $S(Z)$: proxy for the **conditional average treatment effect**: $s_0(Z) = E[Y_1|Z] - E[Y_0|Z]$

$b(Z)$ and $S(Z)$ can be estimated using any supervised learning approach, but we restrict attention to elastic net and random forest (other learners do not increase statistical power)

Testing for Treatment Effect Heterogeneity

Is there heterogeneity in $s_0(Z)$ that can be predicted by $S(Z)$?

- Estimate the regression: $Y = \beta_0 + \beta_1 B(Z) + \beta_2 D + \beta_3 D\left[S(Z) - E[S(Z)]\right]$
  - $\hat{\beta}_2$ provides an (alternate) estimate of the average treatment effect
  - $\hat{\beta}_3$ provides a test of the joint hypothesis that there is treatment effect heterogeneity **and** that treatment effect heterogeneity can be explained by the baseline characteristics

Testing for Differences in Treatment Effects Across Groups

Use $S(Z)$ to partition the sample into quartiles based on predicted treatment effects

- For $k = 1, 2, 3, 4$, let G_k be an indicator equal to one for observations in quartile k
- Define the **sorted group average treatment effect** in quartile k: $\gamma_k = E[s_0(Z)|G_k = 1]$

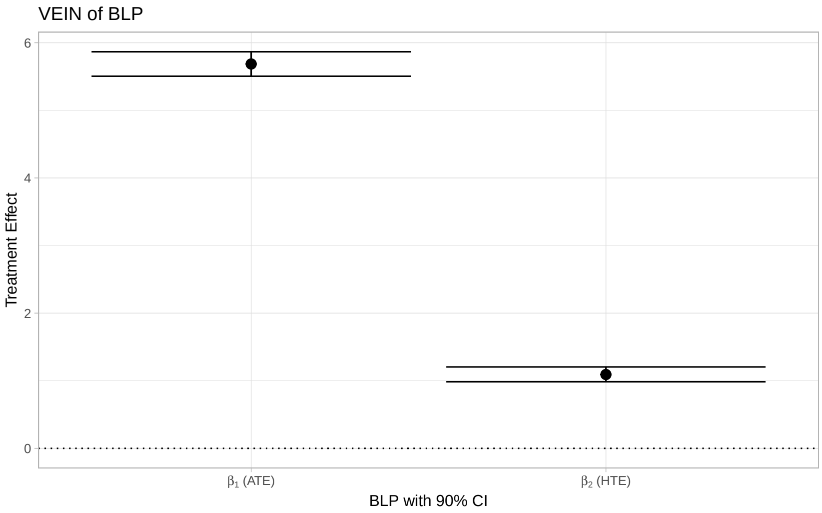
Estimate the regression equation:

$$Y = \theta_0 + \theta_1 B(Z) + \sum_{k=1}^{4} \alpha_k DG_k$$

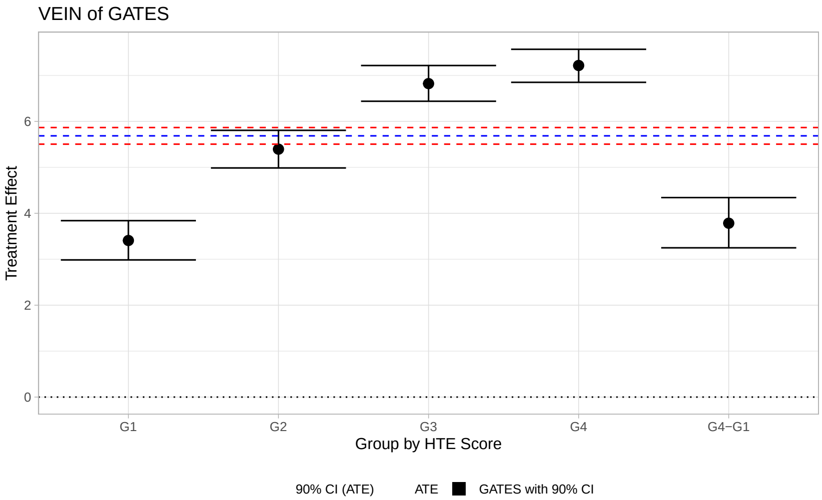
- For $k = 1, 2, 3, 4$, $\alpha_k = \gamma_k$, providing a test of the null hypothesis of no effect on Group k
- Testing the hypothesis that $\alpha_1 = \alpha_4$ is another test of treatment effect heterogeneity

Use a similar approach to test for differences in baseline X variables across the same groups

Testing for Treatment Effect Heterogeneity Using GML



Sorted Group Average Treatment Effects



Characteristics of Most-Impacted and Least-Impacted Groups